FlexBridge: A Flexibility controlled $\mathrm{SE}(3)$ Stochastic Bridge for RNA Ensemble Generation

Anonymous Author(s)
Affiliation
Address
email

Abstract

RNA function is often governed by conformational ensembles rather than a single static structure, yet existing generators either collapse to one conformation or overlook how flexibility varies across nucleotides. We close both gaps with FlexBridge, a simulation-free stochastic bridge on the rigid-motion group $\mathrm{SE}(3)$ that generates RNA conformational ensembles directly from sequence. Diversity arises from independent stochastic trajectories, while a sequence-conditioned, residue-wise flexibility profile, grounded by experimental B-factors, modulates the bridge width to turn heterogeneous nucleotide flexibility into per-residue spatial uncertainty. Empirically, FlexBridge attains strong ensemble coverage of experimentally observed RNA conformations while reproducing the per-nucleotide flexibility pattern that baselines miss.

1 Introduction

RNA molecules are intrinsically dynamic, often interconverting among multiple conformational states rather than occupying a single static structure as it engages with its cellular environment [1–4]. Such ensembles are central to many RNA functions, including ligand recognition, molecular switching, and regulatory control. For example, the preQ$_1$ riboswitch populates multiple ligand-free conformational states, and ligand binding shifts the ensemble toward a functional bound fold [5]. Sequence-conditioned RNA modeling should therefore move beyond predicting one representative structure and instead aim to recover the accessible conformational states and local mobility patterns of RNA ensembles.

Recent deep learning methods have substantially advanced RNA 3D modeling. AlphaFold3 [6] and RhoFold+ [7] achieve strong single-structure accuracy when one representative conformation is desired. Alternative conformations can sometimes be elicited from large structure predictors by repeated inference under perturbed evolutionary contexts, such as MSA clustering or subsampling [8]. However, these procedures are inference-time search heuristics rather than learned sequence-to-ensemble generators, and they depend on repeated model evaluations and informative homologous sequences. Dedicated generative methods have begun to address RNA ensembles more directly: RNAbpFlow [9] generates RNA backbones conditioned on sequence and predicted base-pair information, while DynaRNA [10] models conformational ensembles through coordinate-space diffusion from structural inputs. Despite this progress, existing approaches still lack an explicit mechanism for calibrating how conformational uncertainty should vary across nucleotides.

This missing mechanism matters because RNA uncertainty is not spatially uniform. RNA conformational landscapes are rugged and multi-basin, with metastable states separated by energetic barriers [1, 2]. At the same time, mobility is strongly heterogeneous across structural elements: base-paired helices are relatively constrained, whereas bulges, loops, junctions, and hinge regions can undergo larger-amplitude motion [11, 12, 3]. A generator that injects stochasticity only globally

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faces a precision-recall tradeoff: increasing noise may improve state coverage, but can also allocate unnecessary variance to rigid regions while still under-exploring genuinely flexible ones. In other words, diversity should not be injected uniformly: the generator should allocate uncertainty where the molecule is structurally able to move.

We introduce FlexBridge, a sequence-conditioned stochastic bridge on $\mathrm{SE}(3)^L$ for RNA backbone-frame ensemble generation. FlexBridge uses independent stochastic bridge trajectories to sample multiple conformational basins, while a learned residue-wise width profile $\nu(s)$ makes the bridge heteroscedastic across nucleotides. We ground this profile using chain-normalized experimental B-factor profiles, treating B-factors as noisy but widely available proxies for relative local mobility. This converts residue-level mobility information into spatially varying generative uncertainty: flexible nucleotides receive broader bridge distributions, whereas rigid regions receive narrower ones.

Our main contributions are:

- We formulate RNA sequence-to-ensemble generation as a simulation-free stochastic bridge on $\mathrm{SE}(3)^L$, enabling multi-sample RNA backbone-frame generation through independent stochastic trajectories.
- We introduce a heteroscedastic bridge width controlled by a sequence-conditioned residue-wise flexibility profile and supervised by relative experimental B-factor profiles.
- We evaluate FlexBridge on dynamic RNA benchmarks against multi-seed and MSA-subsampling variants of strong structure predictors, dedicated RNA ensemble generators, and internal ablations, measuring ensemble recall-precision, structural diversity, and residue-wise flexibility agreement.

2 Preliminaries

2.1 RNA Representation and Problem Setup

Given an RNA sequence s of length L, our goal is to learn a sequence-conditioned model $p_\theta(G, \hat{B} \mid s)$ that captures the underlying conformational distribution $q(\cdot \mid s)$ while predicting a residue-wise flexibility profile $\hat{B} \in \mathbb{R}^L_{>0}$. The model should generate conformations $\{G^{(k)}\}_{k=1}^K \sim p_\theta(\cdot \mid s)$ that are individually close to experimentally observed structures and, collectively, cover the multi-modal conformational landscape of the sequence. Following RNA-FrameFlow [13], we represent the backbone of nucleotide i as a rigid frame $G_i = (x_i, R_i) \in \mathrm{SE}(3)$, where $\mathrm{SE}(3) = \mathbb{R}^3 \rtimes SO(3)$ is the special Euclidean group of 3D rigid-body transformations. Here, $x_i \in \mathbb{R}^3$ denotes the position of the $C4'$ atom and $R_i \in SO(3)$ specifies the local orientation defined by the $C3'$–$C4'$–$O4'$ sugar geometry. The remaining non-frame backbone atoms are reconstructed from eight torsion angles $\Phi_i \in SO(2)^8$. The full RNA backbone is therefore written as $G = (G_1, \ldots, G_L) \in \mathrm{SE}(3)^L$.

2.2 $\mathrm{SE}(3)$ Flow Matching

Flow matching [14–16] learns a neural velocity field v_θ that transports a tractable prior p_0 to the data distribution p_{data} by integrating an ordinary differential equation (ODE) along a probability path $\{p_t\}_{t\in[0,1]}$. Given a data sample $G_1 = (x_1, R_1) \sim p_{\mathrm{data}}$, a noise sample $G_0 = (x_0, R_0) \sim p_0(\cdot \mid G_1)$, and a time point $t \sim \mathcal{U}(0,1)$, conditional flow matching on $\mathrm{SE}(3)$ constructs an intermediate state G_t by geodesic interpolation between G_0 and G_1. Differentiating this constant-speed path with respect to t yields closed-form conditional velocity targets

$$u_t^x = \frac{x_1 - x_t}{1-t}, \qquad u_t^R = \frac{\log\big(R_t^\top R_1\big)}{1-t} \in \mathfrak{so}(3), \tag{1}$$

where u_t^x is the remaining translational displacement toward x_1, and u_t^R is the tangent vector pointing from R_t to R_1 along the geodesic on $SO(3)$. The network predicts $(\hat{v}_t^x, \hat{v}_t^R)$ and is trained by tangent-space regression:

$$\mathcal{L}_{\mathrm{SE}(3)}(\theta) = \mathbb{E}_{t,\,G_0,\,G_1}\left[\lambda_x \left\|\hat{v}_t^x - u_t^x\right\|_2^2 + \lambda_R \left\|\hat{v}_t^R - u_t^R\right\|_{\mathrm{F}}^2\right], \tag{2}$$

where $\lambda_x, \lambda_R > 0$ balance the translational and rotational terms, and $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm on $\mathfrak{so}(3)$.

2.3 RNA Flexibility and B-factors

The origin of B-factors depends on the experimental modality [17]. In X-ray crystallography, they are refined as atomic displacement parameters, quantifying the spatial spread of electron density around an atom's mean position [11]. In NMR-derived structures, analogous B-factor-like quantities can be estimated from coordinate dispersion across the deposited structural ensemble, with larger dispersion indicating greater local mobility [18, 19]. Despite these differences in origin, larger B-factors are consistently associated with greater positional uncertainty and, by extension, higher local flexibility. They have therefore been adopted as practical residue-level proxies for mobility in recent computational studies of biomolecular flexibility [20, 21].

3 Methodology

We present **FlexBridge**, a sequence-conditioned stochastic-bridge generator for RNA ensemble generation on $\mathrm{SE}(3)$. The input RNA sequence s is encoded by a frozen, pretrained AF3 Pairformer [6] into single and pair representations $(\mathbf{c}^{(s)}, \mathbf{c}^{(p)})$. From these representations, an IPA-based network v_θ [22] predicts the clean endpoint $(\hat{x}_1, \hat{R}_1)$ from the noisy state (G_t, t), while a lightweight head ψ predicts a positive flexibility profile $\nu(s) \in \mathbb{R}^L_{>0}$ whose global level and residue-wise variation modulate the bridge width. We first define the stochastic bridge on $\mathrm{SE}(3)$ in §3.1, then give the details of FlexBridge in §3.2.

3.1 Stochastic Bridge on $\mathrm{SE}(3)$

Given a sequence s, a stochastic bridge is a stochastic interpolation from a tractable prior p_0 to the empirical structure distribution $q(\cdot \mid s)$. Following the RNA frame representation §2.1, we construct this bridge by combining a translational bridge in $\mathbb{R}^3$ with a rotational bridge on $\mathrm{SO}(3)$.

Translation bridge on $\mathbb{R}^3$. For each residue, the translational component is the closed-form stochastic interpolant [23]

$$x_t = (1-t)\, x_0 + t\, x_1 + \sigma_x \sqrt{t(1-t)}\, \varepsilon^x, \qquad \varepsilon^x \sim \mathcal{N}(0, I_3), \tag{3}$$

where $\sigma_x > 0$ controls the translation width. Conditioned on (x_0, x_1), this is the time-t marginal of the Euclidean Brownian bridge with mean $(1-t)x_0 + tx_1$ and covariance $\sigma_x^2 t(1-t) I_3$.

Rotation bridge on $\mathrm{SO}(3)$. An exact intrinsic Brownian bridge on $\mathrm{SO}(3)$ is inconvenient for simulation-free training: its conditioned dynamics are expressed through the Riemannian heat kernel rather than a simple reparameterized marginal [24, 25]. We therefore propose a tractable Lie-algebra pushforward bridge: lift the relative motion to $\mathfrak{so}(3)$, perturb it in Euclidean coordinates, and map the result back to the group through the exponential map.

Given $R_0, R_1 \in \mathrm{SO}(3)$, the matrix $R_0^\top R_1$ is the relative rotation that moves the frame orientation from R_0 to R_1. We write its Lie-algebra displacement as

$$\Omega_1 = \log(R_0^\top R_1) \in \mathfrak{so}(3), \qquad \omega_1 = \Omega_1^\vee \in \mathbb{R}^3.$$

Here $u^\wedge$ is the skew-symmetric matrix satisfying $u^\wedge v = u \times v$, and $(\cdot)^\vee$ denotes the inverse map. We define the rotational bridge as a Euclidean Brownian bridge in these coordinates, pinned at 0 and ω_1. The Lie-algebra SDE is

$$d\omega_t = \frac{\omega_1 - \omega_t}{1-t}\, dt + \sigma_R\, dW_t, \qquad \omega_0 = 0, \tag{4}$$

for $0 \le t < 1$, where W_t is standard Brownian motion in $\mathbb{R}^3$. Its time-t marginal has the closed-form reparameterization

$$\omega_t = t\omega_1 + \sigma_R \sqrt{t(1-t)}\, \varepsilon^R, \qquad \varepsilon^R \sim \mathcal{N}(0, I_3), \tag{5}$$

and the corresponding rotation is

$$R_t = R_0 \exp(\omega_t^\wedge) \in \mathrm{SO}(3). \tag{6}$$

This construction is group-valued, endpoint-consistent, and simulation-free: $R_t \in \mathrm{SO}(3)$ for all t, $R_{t=0} = R_0$, $R_{t=1} = R_1$, and Eqs. (5)–(6) provide a closed-form reparameterized marginal.

Local approximation to the intrinsic bridge. The Lie-algebra pushforward bridge is not only computationally convenient: it matches the leading local dynamics of the intrinsic bridge. Writing a rotation path in local coordinates as $y_t = \log(R_0^\top R_t)^\vee$, the drift of the intrinsic $\mathrm{SO}(3)$ Brownian bridge from R_0 to R_1 has leading term $(\omega_1 - y_t)/(1-t)$, with curvature-dependent corrections appearing at higher order. Eq. (4) uses exactly this leading term, so the proposed bridge is a first-order approximation to the intrinsic Brownian bridge while retaining a closed-form marginal. Appendix A gives the derivation.

3.2 FlexBridge: a flexibility-scaled bridge

FlexBridge replaces the constant bridge widths (σ_x, σ_R) by widths controlled by a positive flexibility profile $\nu(s)$. The profile is predicted by the lightweight head ψ as

$$\nu(s) = \mathrm{softplus}(\psi(\mathbf{c}^{(s)}, \mathbf{c}^{(p)})) + \epsilon\mathbf{1} \in \mathbb{R}^L_{>0}. \tag{7}$$

Each coordinate $\nu_i(s)$ gives a per-nucleotide flexibility level, so the bridge width varies along the sequence. Define the residue-wise bridge scale $\alpha(t) = \sqrt{\nu(s)t(1-t)} \in \mathbb{R}^L_{>0}$. Substituting the scaled widths into (3) and (5)–(6) gives the FlexBridge corruption

$$\begin{aligned} x_t &= (1-t)x_0 + t\,x_1 + \sigma_x\,\alpha(t) \odot \varepsilon^x, \\ R_t &= R_0 \exp\Big(\big(t\,\omega_1 + \sigma_R\,\alpha(t) \odot \varepsilon^R\big)^\wedge\Big), \end{aligned} \tag{8}$$

Since our Lie-algebra bridge keeps both translational and rotational perturbations in Euclidean coordinates, the per-residue B-factor, itself a Cartesian positional uncertainty in $\mathbb{R}^3$, can directly supervise ν. We regress $\log \nu$ against the log B-factor over residues $\mathcal{V}$ with valid measurements:

$$\mathcal{L}_B = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \big(\log \nu_i - \log B_i\big)^2. \tag{9}$$

Training objective. We train the endpoint network on the flexibility-scaled corruption in (8). For $G_1 \sim q(\cdot \mid s)$, $G_0 \sim p_0$, $t \sim \mathcal{U}(0,1)$, and $G_t \sim p_t^\nu(\cdot \mid G_0, G_1, s)$, the network predicts

$$v_\theta(G_t, t, s) = (\hat{x}_1, \hat{R}_1).$$

The bridge loss is the denoising endpoint-regression surrogate

$$\mathcal{L}_{\mathrm{bridge}}(\theta) = \mathbb{E}\left[\frac{\lambda_x \big\|\hat{x}_1 - x_1\big\|^2_{\mathcal{R}} + \lambda_R \big\|\log(R_t^\top \hat{R}_1)^\vee - \log(R_t^\top R_1)^\vee\big\|^2_{\mathcal{R}}}{(1-t)^2}\right], \tag{10}$$

The $(1-t)^{-2}$ factor makes this endpoint regression equivalent to velocity-space regression up to constants. The total objective is

$$\mathcal{L}_{\mathrm{FlexBridge}} = \mathcal{L}_{\mathrm{bridge}} + \lambda_B\,\mathcal{L}_B + \lambda_\nu\,\mathcal{L}_\nu, \qquad \mathcal{L}_\nu = \frac{1}{L}\|\log \nu\|_2^2, \tag{11}$$

where $\mathcal{L}_{\mathrm{bridge}}$ trains the endpoint network v_θ, $\mathcal{L}_B$ trains the flexibility head ψ against the physical B-factor, and $\mathcal{L}_\nu$ is a log-space anchor that pulls $\log \nu$ toward 0 (i.e., ν toward 1). The anchor both prevents ν from collapsing to an extreme value and provides a meaningful prior for residues without valid B-factor supervision, where $\mathcal{L}_B$ contributes no gradient.

3.3 Inference

At test time, each ensemble member is generated by an N-step Euler–Maruyama integration from $G_0^{(k)} \sim p_0$ with step size $\Delta t = 1/N$. At time $t \in \{0, \Delta t, \ldots, 1 - \Delta t\}$, the endpoint prediction gives the drift

$$\hat{v}_t^x = \frac{\hat{x}_1 - x_t}{1-t}, \qquad \hat{v}_t^R = \frac{\log(R_t^\top \hat{R}_1)^\vee}{1-t},$$

and the residue-wise sampler scale is $\alpha(t) = \sqrt{\nu(s)t(1-t)}$. With independent standard-Gaussian noises $\xi_t^x, \xi_t^R \in \mathbb{R}^{L\times 3}$, applied in translation and Lie-algebra rotation coordinates, we update

$$\begin{aligned} x_{t+\Delta t} &= x_t + \Delta t\,\hat{v}_t^x + \sigma_x\sqrt{\Delta t}\,\alpha(t) \odot \xi_t^x, \\ R_{t+\Delta t} &= R_t \exp\Big(\big(\Delta t\,\hat{v}_t^R + \sigma_R\sqrt{\Delta t}\,\alpha(t) \odot \xi_t^R\big)^\wedge\Big). \end{aligned} \tag{12}$$

The K ensemble members use independent initial states $G_0^{(k)}$ and independent sampler noise trajectories.

4 Experiments

A realistic RNA ensemble generator should recover multiple experimentally observed conformations and assign larger motion to flexible residues. We therefore organize the main experiments around two questions and one ablation. **Q1: Ensemble coverage** tests the central goal of ensemble generation: recovering multiple experimentally observed conformational states. **Q2: Heterogeneous flexibility** examines whether the FlexBridge captures the nucleotide-level heterogeneity.

4.1 Dataset Construction

Our ensemble dataset totals $3{,}218$ structures across $1{,}744$ sequences. Chains are curated from PDB [26] through the BGSU RNA 3D Hub representative set (release 4.30) [27], retaining chains of 20–300 nt with resolution ≤ 4.0 Å. To avoid data leakage, we cluster sequences with CD-HIT at 80% sequence identity and split entire clusters, holding out 10% of the clusters as the test set. Entries released before 2025-01-01 are removed. The final test set contains 29 ensemble examples and 66 single structure examples.

4.2 Baseline Panel

We evaluate against a curated baseline panel. Because RNA ensemble generation is an under-explored setting with few dedicated baselines, we additionally construct new ensemble-generation baselines by adapting two previously single-output predictors (RhoFold+ and AF3) through two orthogonal stochasticity protocols — multi-seed decoding and AFCluster-style MSA subsampling [8] — each reported as a separate row:

- **RhoFold+** [**?**]: a single-structure RNA predictor built on a pretrained RNA language model, adapted to ensemble generation with multi-seed decoding and AFCluster-style MSA subsampling.
- **AlphaFold3** [6]: a diffusion-based biomolecular structure predictor, likewise adapted to ensemble generation with multi-seed decoding and AFCluster-style MSA subsampling.
- **RNAbpFlow** [**?**]: an existing flow-matching generator for RNA backbones conditioned on sequence and predicted base pairs.
- **DynaRNA** [10]: an existing structure-conditioned RNA ensemble generator based on a diffusion model.

All methods produce $K = 60$ conformations per sequence under a matched compute budget.

4.3 Evaluation Metrics

We evaluate generated ensembles from four complementary angles: recall and precision against experimental states, internal ensemble diversity, structural confidence, and residue-wise flexibility agreement with experimental mobility proxies.

Ensemble recall and precision. Let $\mathcal{R} = \{r_k\}_{k=1}^{M}$ be the experimental state representatives for one sequence and $\mathcal{G} = \{g_j\}_{j=1}^{K}$ be generated samples. We report

$$\text{COV-R}_\delta = \frac{1}{M}\sum_{k=1}^{M} \mathbf{1}\left[\min_{g\in\mathcal{G}} \text{RMSD}(g, r_k) \leq \delta\right], \qquad \text{MAT-R} = \frac{1}{M}\sum_{k=1}^{M} \min_{g\in\mathcal{G}} \text{RMSD}(g, r_k),$$

$$\text{COV-P}_\delta = \frac{1}{K}\sum_{j=1}^{K} \mathbf{1}\left[\min_{r\in\mathcal{R}} \text{RMSD}(g_j, r) \leq \delta\right], \qquad \text{MAT-P} = \frac{1}{K}\sum_{j=1}^{K} \min_{r\in\mathcal{R}} \text{RMSD}(g_j, r).$$

Recall quantifies how many observed states are recovered by the generated ensemble, whereas precision measures whether generated samples remain close to at least one observed state.

Diversity. We measure how broadly a method explores conformational space by the mean pairwise $C4'$ RMSD within each generated ensemble,

$$\mathrm{Div}(\mathcal{G}) = \binom{K}{2}^{-1} \sum_{1 \le i < j \le K} \mathrm{RMSD}(g_i, g_j), \tag{13}$$

which collapses toward zero when the ensemble concentrates on a single conformation.

Confidence. To guard against methods that trade accuracy for diversity by producing unrealistic conformations, we score the structural plausibility of each generated sample with LocIParse [28], a locality-aware invariant point attention model for RNA 3D structures. For an ensemble $\mathcal{G}$, we report the mean LocIParse score

$$\mathrm{Conf}(\mathcal{G}) = \frac{1}{K} \sum_{j=1}^{K} \mathrm{LocIParse}(g_j).$$

4.4 Ensemble Coverage and Sample Quality

We evaluate how well each generator recovers the experimental conformational ensemble. Table 1 reports recall and precision against experimental states, internal diversity, and LocIParse confidence for the baseline panel and FlexBridge.

Table 1: Ensemble coverage and sample quality on the ensemble test set (43 sequences, $K = 60$).

	Recall to experimental states			Precision to experimental states			Diversity	Confidence
Method	COV-R@2 ↑	COV-R@4 ↑	MAT-R ↓	COV-P@2 ↑	COV-P@4 ↑	MAT-P ↓	Pairwise RMSD ↑	Conf ↑
RhoFold+ (multi-seed)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RhoFold+ (MSA subsampling)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
AF3 (multi-seed)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
AF3 (MSA subsampling)	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RNAbpFlow	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DynaRNA	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
FlexBridge	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

4.5 Heterogeneous Flexibility

Table 2: Heterogeneous flexibility on the ensemble test set. $\mathcal{L}(\nu, B)$ tests whether our flexibility predictor is accurate; $\rho(\nu, \hat{\sigma}^2)$ tests whether ν controls the generation process; $\rho(\hat{\sigma}^2, B)$ tests whether the generated samples are consistent with physics.

Method	$\mathcal{L}(\nu_i, B_i) \downarrow$	$\rho(\nu_i, \hat{\sigma}_i^2) \uparrow$	$\rho(\hat{\sigma}_i^2, B_i) \uparrow$
AlphaFold3	N/A	N/A	[TODO:]
RNAbpFlow	N/A	N/A	[TODO:]
DynaRNA	N/A	N/A	[TODO:]
Deterministic SE(3) flow	N/A	N/A	[TODO:]
EnsBridge ($\nu \equiv 1$) (ours)	N/A	N/A	[TODO:]
FlexBridge w/o $\mathcal{L}_B$ (ours)	N/A	N/A	[TODO:]
FlexBridge (ours)	[TODO:]	[TODO:]	[TODO:]

4.6 Ablation on Ensemble Coverage

Table 3: Ablation on ensemble coverage.

Variant	COV-R@2 ↑	COV-R@4 ↑	MAT-R ↓	Pairwise RMSD ↑
EnsBridge ($\nu \equiv 1$) (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
FlexBridge w/o $\mathcal{L}_B$ (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
Deterministic $\mathrm{SE}(3)$ flow	[TODO:]	[TODO:]	[TODO:]	[TODO:]
FlexBridge (ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

5 Related Work

TBD...

6 Conclusion

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A Local Approximation of the Rotation Bridge

B Single-Conformation Accuracy

Table 4: Single-conformation accuracy on the single-structure test set.

	$C4'$ RMSD			TM-score		
Method	Best-of-1 ↓	Best-of-20 ↓	Best-of-50 ↓	Best-of-1 ↑	Best-of-20 ↑	Best-of-50 ↑
RhoFold+	[TODO:]	–	–	[TODO:]	–	–
RoseTTAFoldNA	[TODO:]	–	–	[TODO:]	–	–
AlphaFold3	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
RNAbpFlow	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
DynaRNA	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
FlexBridge	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]